What Are We Measuring? A Benchmark Audit and a Failed Zone Hypothesis in Emotion Recognition for Autistic Children

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Abstract—**Emotion recognition is widely proposed as assistive technology for autistic children, and papers on public autism facial-expression archives report up to 97.95% accuracy. We audit the archive behind many of those figures. It ships two overlapping dataset trees sharing 45% of the smaller tree's images, 21.4% of its files are byte-identical duplicates, and 103 duplicate groups carry contradictory labels on pixel-identical images. Holding images, features, and classifier fixed and varying only the split protocol, five-category balanced accuracy spans 0.346–0.750; duplicate leakage explains little of this, worth 5.0 points and 2.1 to a nearest-neighbour memoriser. The contradictions have a sharp geometry: one boundary, valence, absorbs 102 of 103, and a permutation null puts cross-valence conflict at a roughly 200-fold depletion. Controls then refute our own prior three-zone framework—it absorbs none of the contradictions, because it cuts between anger and sadness where disagreement peaks, and its apparent stability advantage is generic to class count. A binary valence target reaches 0.890 under a group-aware split against 0.727 for three zones, so this archive supports a valence decision and nothing finer. This is a measurement audit, not a deployable classifier, and not a claim about any child's internal state.**

Index Terms—**benchmark integrity, data leakage, label noise, affective computing, autism**

I. Introduction

Roughly 1 in 31 eight-year-olds in the United States is diagnosed with autism spectrum disorder (ASD) [1]. Because emotional expression in autistic people is frequently misread by non-autistic observers—the *double empathy problem* [2]—automated emotion recognition is repeatedly proposed as a communication aid.

The literature on this task is in an unusual state. Papers evaluating on public autism facial-expression archives report accuracies from 80.0% [3] to 97.95% [4]. Papers evaluating on independently collected, consented clinical corpora mostly report less: 40% for a commercial analyser on the EMBOA autism corpus [5] and 57.95–64.92% for four commercial classifiers on parent-reported autistic children [6]. The comparison needs care. Grossard *et al.* [7] reach 82.05% within-population on lab-collected video, inside the public-archive band, and 66.43% training on typically developing children and testing on autistic ones—so part of any gap is population transfer, not data provenance. What remains unexplained is the top of the public-archive band, which no consented corpus approaches.

That correlation has become hard to ignore. Publishers have retracted work built on scraped autism face datasets, including the source paper for the archive we audit [8] and a facial emotion recognition paper [9], on the grounds that images were collected from the internet without documented diagnosis, ethical oversight, or guardian consent. That established a *provenance* failure. It did not establish whether the reported numbers were also measurement artifacts.

This paper supplies that measurement: an integrity audit of the archive as shipped (Sec. III); a protocol ablation with a nearest-neighbour reference memoriser (Sec. IV); a partition analysis locating the label contradictions, with a permutation null (Sec. V); and a control experiment that refutes our own three-zone hypothesis, leaving a narrower claim that survives it (Sec. VI). We audit an instrument rather than build one, make no claim about any child's internal state, and treat every label as a filing decision about an image rather than ground truth about a person.

II. Related Work

Cross-population transfer is established. Grossard *et al.* [7] ran the controlled experiment (157 typically developing and 36 autistic children, subject-independent CV), and Gaya-Morey *et al.* [10] repeated the design with twelve CNNs on an adjacent population. We do not claim that finding; we ask whether the public benchmarks now used to measure it can support the measurement.

Benchmark contamination is a mature field. Kapoor and Narayanan [11] catalogue eight leakage types across 294 papers; Barz and Denzler [12] found 3.3% and 10% duplicate rates in CIFAR test sets; Northcutt *et al.* [13] showed label errors reorder rankings, and Recht *et al.* [14] that the evaluation set alone moves accuracy. We adopt the taxonomy of Ramos *et al.* [15]—*hard* leakage for identical images across a split, *soft* for near duplicates, *intra-dataset* for overlap within one

TABLE I

TWO DATASET TREES SHIPPED IN ONE ARCHIVE.

Tree	Files	Unique	Train	Test
Balanced copy	1,425	1,380	1,199	226
Imbalanced copy	833	717	758	75
Shared unique images: 323 (45.0% of smaller tree)				

dataset's splits—and the notions of *augmented duplicates* [16] and *synthesis leakage* [17], where train and test items derive from a common source under a transform. None of this work examines a clinical affect dataset.

Contamination in this archive has been asserted, not measured. The authors of the FERAC derivative [18] reported "class overlap and image duplication" here, removed duplicates, and dropped two classes as unrecoverable. They reported no detection method, threshold, duplicate rate, leakage rate, or effect of cleaning. Two groups have now flagged the same archive; what remains open is the quantity.

Dataset critique. Birhane and Prabhu [19] argue scraped corpora containing people, including minors, are ethically defective at collection; Gebru *et al.* [20] propose the dataset documentation practices absent here.

Why the labels might be the problem. Barrett *et al.* [21] challenge the premise that emotional state is readable from facial configuration at all. Keating *et al.* [22], from over 265 million motion-capture points, find autistic and non-autistic expressions differ in which muscles carry each emotion. If expressions are ambiguous under a non-autistic reading, forcing them into mutually exclusive categories should produce contradictions concentrated where categories are closest.

III. THE ARCHIVE

We audit the public Kaggle archive associated with [8].[1] The 1,429 distinct SHA-256 digests covering the 1,672 files retained by an earlier run of ours—which excluded `surprise` and the conflicted groups—are all present in the fresh download, confirming the archive has not changed under us. The audit here covers all 1,774 distinct digests. Statistics below come from the vendor's own directory layout, before any resplitting: this is what a downstream user receives.

Two overlapping datasets ship as one. The archive contains two top-level trees (Table I) with different class balances and splits, sharing 323 unique images—45.0% of the smaller. This plausibly explains why published descriptions of this dataset report its size variously as 348, 830, 1,603, and "2500+": different papers loaded different trees.

[1]Slug `fatmamtalaat/autistic-children--emotions-dr-fatma-m-talaat`. We name it for reproducibility while noting it distributes images of minors of disputed provenance; readers should weigh that before downloading.

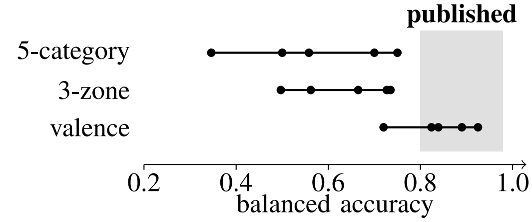

Fig. 1. Each dot is one split protocol over the same images, features, and classifier; test sets differ in size, so part of each spread (40.4, 23.8, 20.5 points) is sampling noise. Under five categories the best result, 0.755 (SE ≈0.03), sits 4.5 points below the published band's lower edge and 22 below its upper. Only binary valence enters the band, because the question is easier—not because the archive is cleaner.

One file in five is a duplicate. Of 2,258 labelled images only 1,774 are distinct by SHA-256; 484 files (21.4%) are byte-identical copies, in groups reaching size 5. Perceptual hashing at Hamming distance 4 adds 163 near-duplicate components. A filename suffix (`_aug_N`) identifies 1,020 augmentation families, 335 larger than one, so part of the archive is synthetic variation on fewer source images—synthesis leakage [17], and identifiable from filenames alone without any image analysis.

Pixel-identical images carry contradictory labels. Under the vendor's six categories, 119 byte-identical groups (327 files) are filed under more than one emotion. Establishing this needs no external ground truth and no annotator panel, only a hash—an internal inconsistency of the artifact, which is cheaper and stronger than label-error detection [13].

IV. HOW MUCH OF THE NUMBER IS THE PROTOCOL?

Contamination matters only if it moves results. We hold the images, the features, and the classifier fixed and vary *only* how train and test are separated. Features are frozen `ViT-B/16` embeddings [23]; the head is a one-hidden-layer MLP with epoch-wise early stopping; every protocol runs over ten seeds [24]. Because `surprise` has no zone, it is dropped from *both* arms, so the categorical arm is a five-way task over exactly the images the zone arm sees and the two columns differ only in label granularity.

We add a 1-nearest-neighbour classifier as a *reference memoriser*. It stores the training set exactly, so a test image whose byte-identical twin is in training is recovered with certainty. It is not an upper bound over all models—a fine-tuned network memorises and generalises, and we did not run one—but it is the estimator most able to convert an exact duplicate into a correct prediction, which makes it the sharpest available probe of how much these duplicates are worth under this representation.

Table II and Fig. 1 support three claims and refute a fourth—that leakage explains the published figures—which we expected to make.

The reported number is largely a choice. Across five defensible protocols on identical images, five-category bal-

TABLE II
SAME IMAGE POOL, FEATURES, AND CLASSIFIER; ONLY THE SPLIT CHANGES—SO THE TEST SETS DIFFER IN SIZE AND COMPOSITION, AND n IS GIVEN FOR EACH. "GRP" IS THE SHARE OF TEST IMAGES WHOSE DUPLICATE OR AUGMENTATION FAMILY ALSO APPEARS IN TRAINING. TEN SEEDS; SD IS OVER SEEDS AND DOES NOT INCLUDE TEST-SET SAMPLING ERROR, WHICH AT $n = 69$ IS ROUGHLY ± 0.06.

Protocol	Bal. acc.	sd	1-NN	Grp	n
Five categories (chance = 0.200)					
Vendor split, imbal. tree	0.346	.050	0.348	26.1%	69
Train one tree, test other	0.500	.043	0.476	24.6%	69
Vendor split, bal. tree	0.558	.037	0.585	33.3%	189
Group-aware	0.700	.052	0.734	0.0%	180
Pooled random	0.750	.028	0.755	80.6%	293
Three zones (chance = 0.333)					
Vendor split, imbal. tree	0.497	.063	0.392	26.1%	69
Vendor split, bal. tree	0.562	.025	0.552	33.3%	189
Train one tree, test other	0.665	.025	0.661	24.6%	69
Group-aware	0.727	.034	0.733	0.0%	180
Pooled random	0.735	.023	0.736	80.6%	293
Binary valence (chance = 0.500)					
Vendor split, imbal. tree	0.720	.011	0.569	26.1%	69
Vendor split, bal. tree	0.824	.029	0.797	33.3%	189
Train one tree, test other	0.839	.030	0.858	24.6%	69
Group-aware	0.890	.019	0.869	0.0%	180
Pooled random	0.925	.017	0.926	80.6%	293

anced accuracy spans 0.346 to 0.750, a 40.4-point range. The protocols also differ in test-set size and composition, so this is protocol plus sampling rather than protocol alone; the smallest cells hold 69 images. Seed dispersion is 2.3–6.3 points, several times smaller than the protocol spread even after allowing that the range of five draws is itself about 2.3σ, and test-set sampling error at $n = 69$ is roughly ± 0.06. A reader given one number cannot tell where in that band it sits.

The zone collapse narrows the spread. The same protocols span 23.8 points under three zones against 40.4 under five categories. Because the label spaces have different chance floors we compare them chance-corrected as $(\mathrm{BA} - 1/k)/(1 - 1/k)$. Sec. VI quantifies the reduction under a single fixed split, where the controls also run, and shows it is not a property of *these* classes.

Duplicate leakage is real but second-order. Pooled random leaks the family of 80.6% of its test images and exceeds the group-aware split by 5.0 points under five categories and 0.8 under zones. The reference memoriser gains even less: 1-NN improves by only 2.1 and 0.3 points from the same contamination, so the duplicates are worth little even to the estimator best placed to exploit them. 1-NN also matches or beats the trained head on three of five five-category protocols,

TABLE III
BYTE-IDENTICAL DUPLICATE GROUPS STILL CARRYING CONFLICTING LABELS AFTER COARSENING, AGAINST 346 DUPLICATE GROUPS AND A 103-CONFLICT FIVE-WAY BASELINE.

Coarsening	Conflicts	Absorbed
Five categories (baseline)	103	—
{natural, joy} vs rest **(valence)**	**1**	**102**
{anger, sadness} vs rest	26	77
{anger} vs rest	77	26
{anger, fear} vs rest	102	1
Valence zones {nat,joy}{ang,fear}{sad}	103	0
Best of 15 three-block partitions	26	77

which bounds how much the head is extracting beyond nearest-neighbour structure and is a further reason to read these absolute values as a floor.

This form of contamination does not reach the published band. Under five categories the best value any protocol reaches, with any estimator, is 0.755—below the 80.0–97.95% band reported on archives of this family [3], [4]. So *exact-duplicate and augmentation-family leakage, under frozen* `ViT-B/16` *features and our re-splits*, does not by itself produce those figures. This is not a claim that contamination is irrelevant: augmentation applied before splitting in downstream derivatives is itself synthesis leakage [17] and would not appear in our accounting, since we audit the archive as shipped. Small test sets, selection among runs, and fine-tuned networks on contaminated splits remain untested candidates.

V. WHERE THE CONTRADICTIONS LIVE

The contradictions are not spread evenly across the label space, and locating them turns out to matter more than counting them.

Restrict to the 1,952 images whose category maps to a valence zone (`surprise` has none) and ask, for each way of coarsening those five categories, how many byte-identical duplicate groups still straddle a block boundary. A coarsening that absorbs many contradictions is one whose boundaries the filing process respected. Because every candidate below has the same block shape, this measures *where* the boundary falls, not how many classes there are. Table III gives the result.

One boundary absorbs almost everything. Splitting positive/neutral from negative leaves one conflicted group out of 346; every other binary cut leaves far more. A permutation null preserving the zone marginals and duplicate-group structure sharpens this: it expects 536 conflicted files, 217 touching positive/neutral, where the archive has 280 of which exactly one does. Cross-valence conflict is therefore depleted roughly 200-fold. Conditioning on the 280 conflicts that actually occur—the null over-predicts the total by 1.9×—within-negative conflict is *enriched* about 1.7-fold rather than depleted. The filing process separated valence almost perfectly and anger, fear, and sadness not at all.

The three-zone partition absorbs none of it. This is a negative result about our own prior framework. Valence zones place anger and fear together and sadness apart, but anger versus sadness is the most common contradiction in the archive, so the zone-2/zone-3 boundary runs straight through the region of maximum disagreement. Among the 15 same-shape partitions the zone partition ties for worst at absorbing contradictions, while grouping anger with sadness absorbs 77 of 103.

We do not call this annotator disagreement: the archive is a scrape with no documented labelling procedure, and one image retrieved under two queries and written into two folders explains it as well. Either way the consequence for a model is identical, and it is the consequence our evidence supports.

VI. What This Does and Does Not License

A stability argument for zones does not survive its control. Collapsing five classes to three narrows the chance-corrected protocol range from 0.482 to 0.404, a 16.1% reduction. That reads as support for zones until the control is run. Across all 15 partitions of the same 2-2-1 shape the chance-corrected range runs from 0.325 to 0.628; the zone partition ranks 5th, and a deliberately valence-crossing partition does *better* (0.391, an 18.7% reduction). Fewer classes buys stability; *these* classes buy nothing extra. We report this because we expected the opposite, and because a coarsening cannot be justified by a property that every coarsening shares.

What survives is a granularity claim, and it is sharper. The evidence in Sec. V is specific to one boundary: the reproducible distinction is valence, and the anger/fear/sadness distinctions are not reproducible at all. Table II bears this out where it matters. Under the group-aware split the binary valence target reaches 0.890, against 0.727 for three zones and 0.700 for five categories; the gain is concentrated entirely at the boundary the conflict analysis identifies as reliable. So the archive supports a valence decision well and finer decisions poorly, which argues for a binary target rather than the three-zone scheme we and others have used—or, if three levels are wanted for intervention, for a partition that does not cut between anger and sadness. Note that binary valence does not reduce protocol sensitivity either (chance-corrected range 0.410, against 0.404 for zones): coarsening fixes *what* is being asked, not how much the answer depends on the split.

We stop short of the actionability claim: zones are usually motivated by what a caregiver would do, we ran no caregiver study, and nothing here shows anger and fear share a response. We can say only which distinctions the data support.

Context beyond the face. Even under our strictest split this task saturates well below deployment quality—the group-aware five-category result is 0.700, and because our groups are duplicate families rather than children that is an upper bound on cleanliness. Meanwhile Barrett *et al.* [21] argue facial configuration underdetermines emotional state in principle. If the face reliably yields only valence, then everything clinically interesting—which negative state, and why—must come from elsewhere.

This is where multimodal and vision-language approaches earn their place, on a data argument rather than an accuracy one. A face-only autism-specific model needs autism-specific face data—precisely the resource whose acquisition produced the retractions cited above. A vision-language model brings prior knowledge of scenes, activities, and posture without any autism-specific corpus, and supplies the context a cropped face cannot. Reporting a coarse state with an explanation a caregiver can check is also more auditable than a fine-grained category with a confidence score, when those categories are not reproducible.

VII. Limitations and Ethics

This audits one archive and does not establish prevalence, though [18] reports consistent qualitative findings. Duplicate detection is hash-based and will miss re-encoded or cropped duplicates, so our rates are lower bounds; the Hamming-4 threshold was not validated against hand-labelled pairs. The archive supplies no subject identities, so our groups are duplicate and augmentation families, not people, and a subject-disjoint split would be stricter. Our features come from a web-pretrained backbone, so we cannot exclude that the encoder has seen these images—a caveat applying to every published result here, which our "0% leakage" rows do not escape. The reference memoriser characterises contamination under *these* features only; it is not an upper bound over models, and no fine-tuned network was run.

Ethics. This is secondary analysis of an already-public dataset, with no interaction with human subjects and no attempt to re-identify anyone. We did not seek an institutional determination and do not assert one here. We report only aggregate statistics: no image, filename, hash, or per-image prediction is released. The image corpus was deleted after hashing and feature extraction. We make no claim that any depicted child is autistic, that any label reflects a felt emotion, or that any system should be deployed on this basis; where Sec. V references findings about autistic expression, it does so to motivate a hypothesis, not to assert that this archive depicts diagnosed children. The archive's provenance is disputed and its license unstated, so we redistribute nothing.

VIII. Conclusion

A widely used public autism emotion archive ships two overlapping copies sharing 45% of the smaller, is 21.4% duplicated, and contains 103 groups of pixel-identical images with contradictory labels. Varying only the split protocol moves five-category balanced accuracy across a 40.4-point band, so a single number on this archive says little about a model. Exact-duplicate and augmentation-family leakage is not the main driver—worth 5.0 points, and 2.1 to a reference memoriser—and no protocol we ran reaches the published band, though augmentation applied before splitting in downstream derivatives would not show up in our accounting.

One boundary carries the label structure: valence absorbs 102 of 103 contradictions, and cross-valence conflict is depleted roughly 200-fold against a marginal-preserving null. Our own three-zone framework absorbs none of them, because it cuts between anger and sadness where disagreement is greatest, and its protocol-stability advantage disappears against same-shape controls. The granularity this archive supports is binary valence; anything finer needs context beyond the face. Any future benchmark here should report a duplicate audit, a group-aware split, and chance-corrected metrics as a condition of being believed—and, given how this archive was assembled, should not be built by scraping.

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